

CS101 - Introduction to Computer Science

Midterm Exam

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Section: 002

Instructions: Answer all questions. Show your work where applicable.

Q1. (10 pts) Explain the difference between a stack and a queue. Give a real-world example of each.

Stack is LIFO, queue is FIFO.

Stack example: undo button in editor.

Queue example: printer queue.

Q2. (10 pts) What is the time complexity of binary search? Explain why.

$O(\log n)$ because it divides array in half each time.

Q3. (10 pts) Define recursion and write a recursive function to calculate factorial of n.

Recursion means calling itself.

$\text{factorial}(n) = n * \text{factorial}(n-1)$

base case when $n=1$ return 1
